IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-60

Charging Stations-III

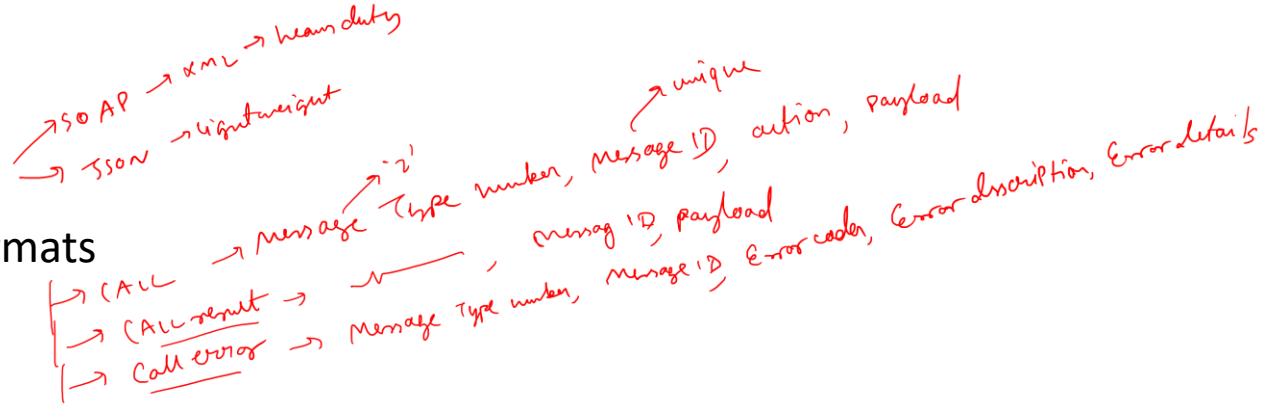
Dr. Apurv Kumar Yadav

Department of Electrical Engineering



Recap

- Charging stations
- OCPP protocol
- OCPP Implementation
- OCPP 1.6J message formats



Boot Notification Request (CALL)

```
[2,
 "12345678",
 "BootNotification",
 { "chargePointVendor": "vendorx",
   "chargePointModel": "SingleCharger"
 }
]
```

(2 → Message Type ID; "12345678" → Message unique ID; "BootNotification" → action; payload)

Boot Notification confirm (CALL Result) or response

```
[3,
 "12345678",
 { "Status": "Accepted",
   "currentTime": "2025-08-15T10:00:30.4",
   "heartbeatInterval": 100
 }
]
```

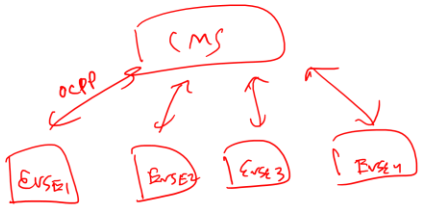
(3 → Message Type ID; "12345678" → Message ID (same as Call message ID))

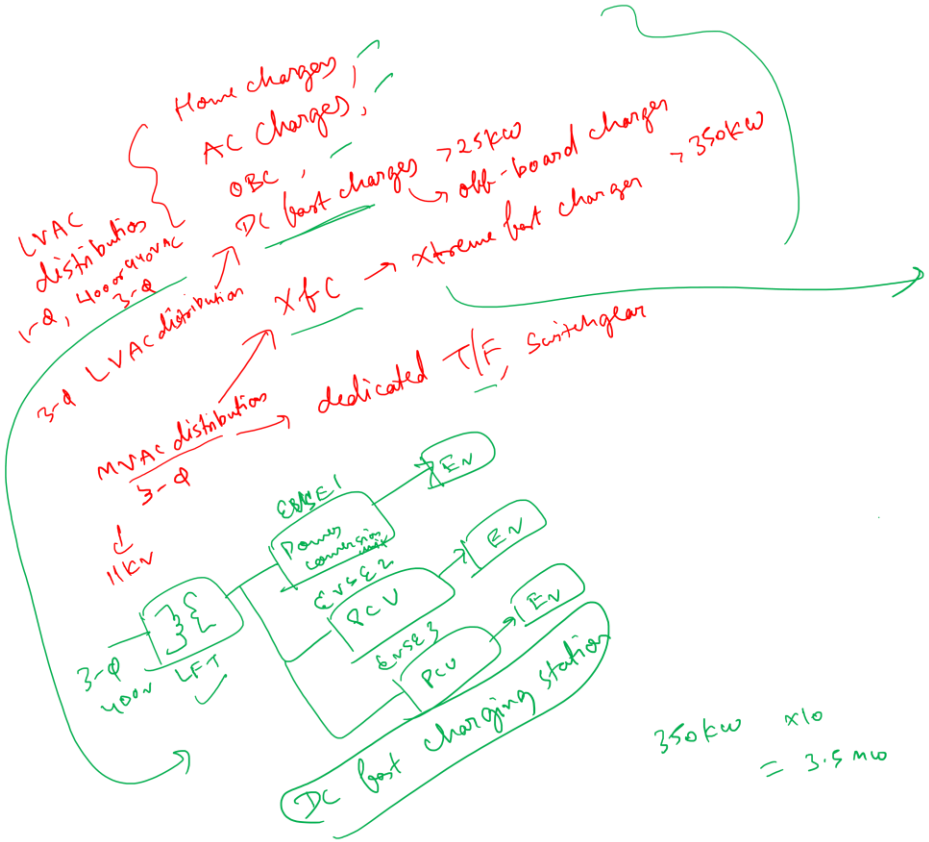
OCPP Profiles

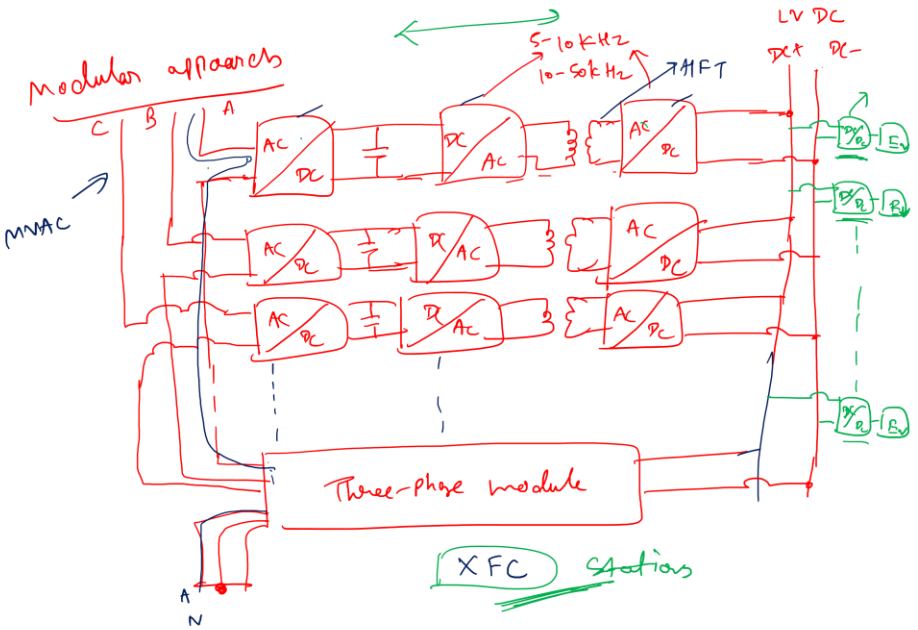
- Many OCPP profile can be added, however the core certification profile must always be present.
- Core certification profile include: Basic Authentication, Security Event Notification, Booting a Charging Station, Configuring a Charging Station, Resetting a Charging Station / EVSE, Stop Transaction, Local start transaction - Cable plugin first & Authorization first, Check Transaction status, Remote start / stop transaction, Remote unlock Connector, Remote Trigger, Change Availability - Charging Station / EVSE / Connector, Clock-aligned Meter & Sampled Meter Values, Retrieve Log Information, Get / Clear Customer Information, Secure Firmware Update, Store / Clear Authorization Data in Authorization Cache

OCPP 2.0.1 Functional Blocks

- There are 16 functional block in OCPP 2.0.1
  - Security, Provisioning, Authorization, Local Authorization List Management, Transactions, Remote Control, Availability, Reservation, Tariff and Cost, Metering, Smart Charging, Firmware Management, Certificate Management, Diagnostics, Display Message, Data Transfer, Bidirectional Power Transfer, DER Control, Battery Swapping

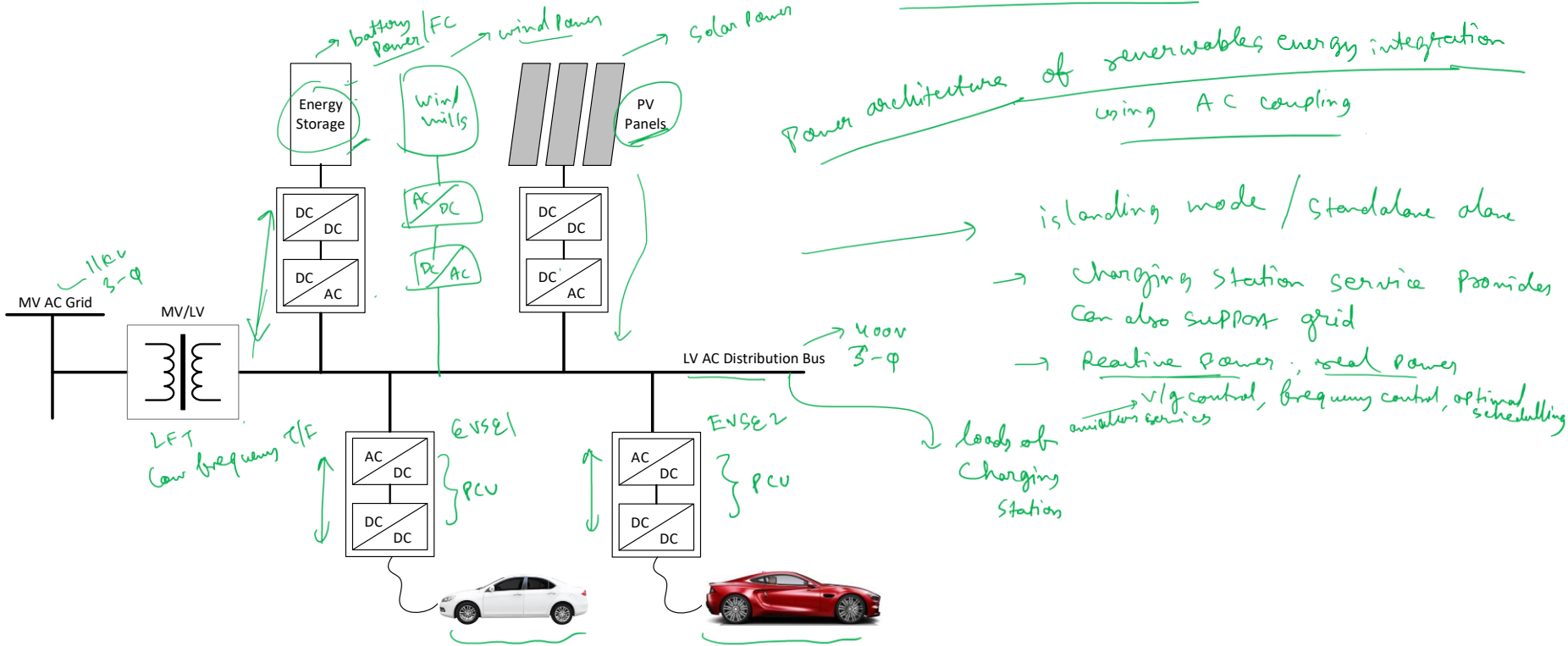






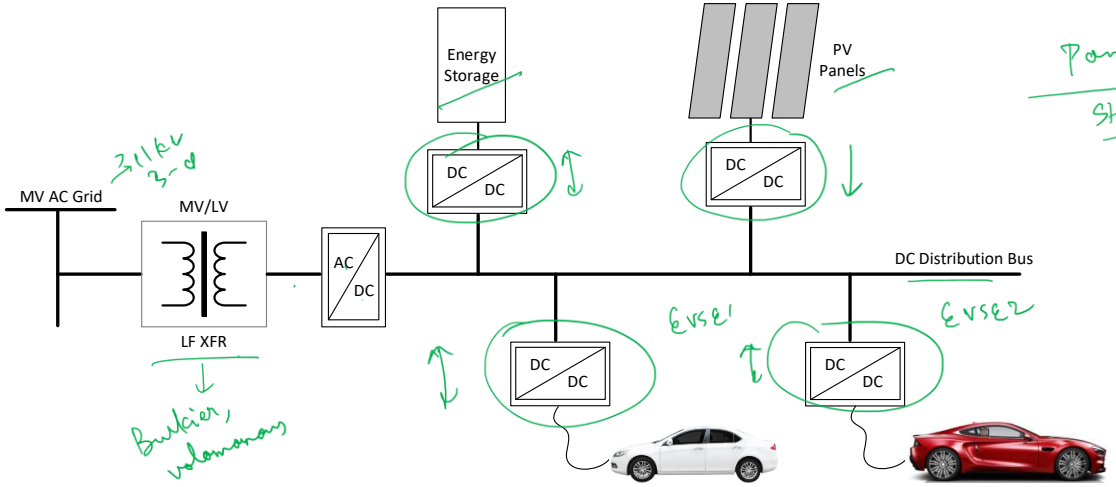
Source: Iqbal Hussain, "Electric and Hybrid Vehicles : Design Fundamentals", 3rd Ed. CRC Press, 2021

High Power DC Charging Station with AC coupling



Source: Iqbal Hussain, "Electric and Hybrid Vehicles : Design Fundamentals", 3rd Ed. CRC Press, 2021

High Power DC Charging Station with DC coupling

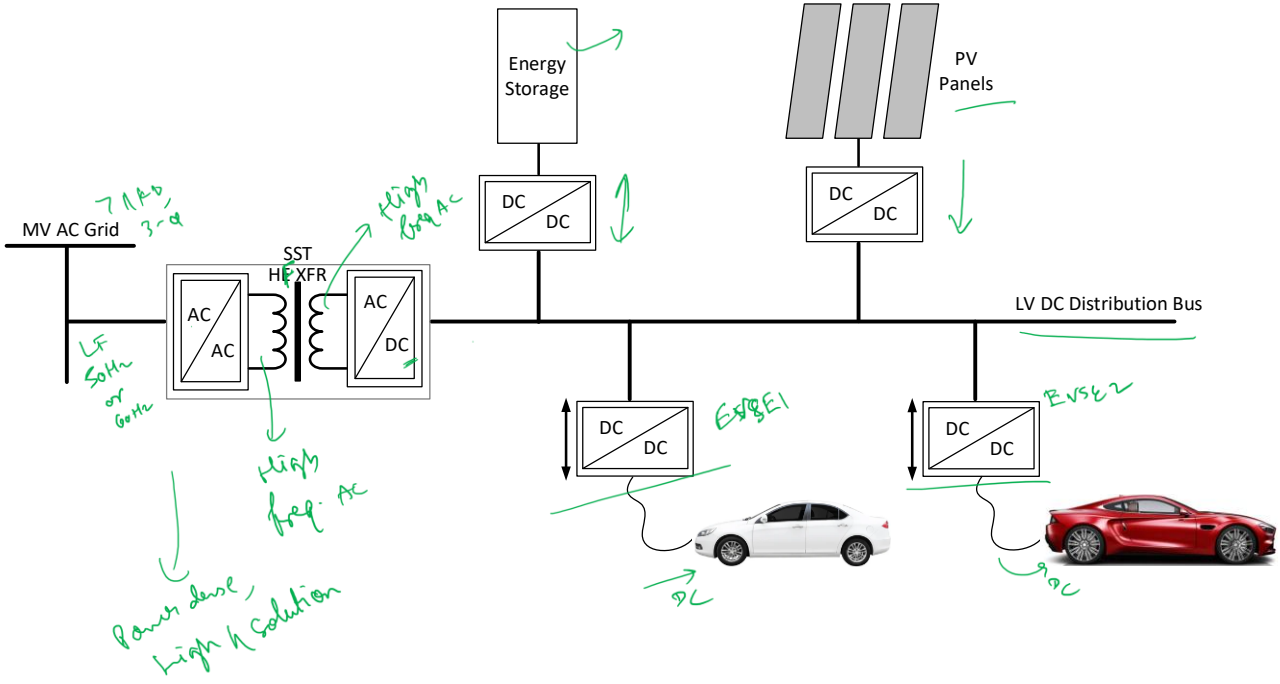


Power architecture of integrating renewables & energy storage system using DC coupling
→ eliminate different AC-DC converters

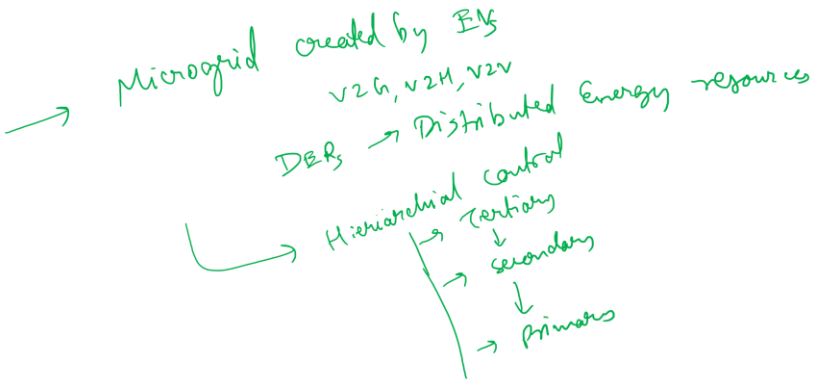
EVSE1
EVSE2
→ optimal loading

Source: Iqbal Hussain, "Electric and Hybrid Vehicles : Design Fundamentals", 3rd Ed. CRC Press, 2021

High Power DC Charging Station with DC coupling



Source: Iqbal Hussain, "Electric and Hybrid Vehicles : Design Fundamentals", 3rd Ed. CRC Press, 2021


Microgrid created by EVs
V2G, V2H, V2V
DERs → Distributed Energy resources
Hierarchical control
tertiary
secondary
primary

Thank You